TCA Report

1. Steps Performed

a. Data Loading and Preprocessing

- Imported the trading dataset (project2_trading.csv) and instrument metadata (project2_instruments.csv).
- Computed additional cost metrics:

$$\text{Slippage in USD} = \frac{\text{slippage_bps}}{10000} \times \text{execution_price} \times \text{quantity}$$

$$\text{Market Impact in USD} = \frac{\text{market_impact_bps}}{10000} \times \text{execution_price} \times \text{quantity}$$

$$\text{Total Trading Cost in USD} = \text{slippage_usd} + \text{commission_usd}$$

b. Merging with Instrument Data

- Merged with instrument metadata to bring in **asset class** and **sector**, allowing cost analysis by category.

c. TCA-Focused Dataset Creation

- Extracted relevant columns: timestamp, strategy, asset_class, sector, slippage (bps & USD), market impact, commissions, total trading cost, realized and unrealized PNL.
- This streamlined dataset formed the basis for further analysis.

d. Descriptive Statistics

- Generated statistical summaries (mean, min, max, quartiles) for all TCA metrics.

2. Sector-Level Insights

- **Best Sector**:
  Technology – Delivered the highest **average P&L** with relatively low total trading costs. Efficient execution in liquid, high-volume instruments.
- **High-Cost Sector**:
  Healthcare – Exhibited the highest **slippage and total trading costs**, indicating challenges in execution due to lower liquidity or volatile conditions.
- **Balanced Sector**:
  Financial – Showed moderate costs and stable P&L, suggesting reliable but not outstanding performance.

3. Asset Class-Level Insights

- **Best Performing**:
  Derivatives – Highest profitability with relatively low trading costs. Strong liquidity helped minimize slippage and impact.

- **Costly Class**:
  Fixed Income – Notable for high market impact and higher overall costs, consistent with bond market illiquidity.

- **Mixed Performance**:
  Crypto – Higher slippage, but select trades offered good opportunities, reflecting sensitivity to contract type and market timing.

4. Strategy Insights

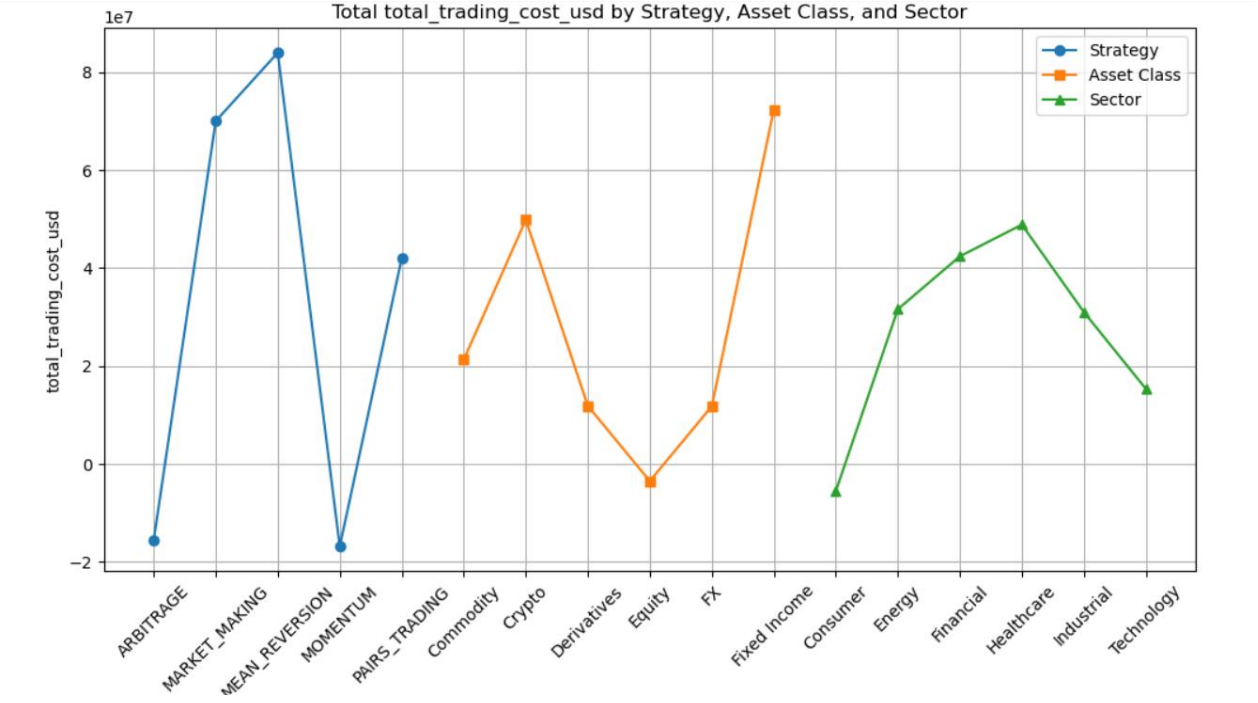
- **Best Strategy**:
  Arbitrage – Generated the most consistent profits with **low slippage and market impact**, especially in Crypto and Consumer assets. Efficient execution made it a reliable profit driver.

- **High-Cost Strategy**:
  Market Making – Carried **higher trading costs** due to constant liquidity provision. While it earned profits in Technology assets, the exposure to higher transaction costs reduced efficiency.

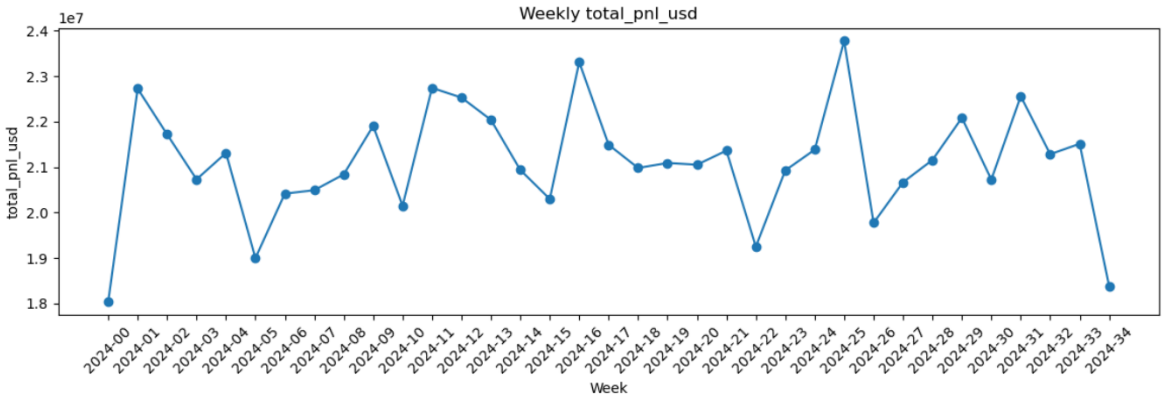
- **High-Risk, High-Reward Strategy**:
  Momentum – Delivered **sharp profits in trending markets** (notably in Crypto and Technology), but also suffered **losses in range-bound periods** due to higher slippage variability.

- **Stable but Modest Strategy**:
  Pairs Trading – Produced **steady but modest P&L** with moderate costs. Worked best in Financials and Energy sectors where spreads could be exploited.

- **Situational Strategy**:
  Mean Reversion – Showed **mixed performance**, profitable in calm markets (Healthcare, FX) but struggled during strong trends. Costs were relatively higher due to going against short-term market moves.




Total pnl_usd by Strategy, Asset Class, and Sector
1e8
pnl_usd
1.5
1.4
1.3
1.2
1.1
Strategy
Asset Class
Sector
ARBITRAGE
MARKET_MAKING
MEAN_REVERSION
MOMENTUM
PAIRS_TRADING
Commodity
Crypto
Derivatives
Equity
FX
Fixed Income
Consumer
Energy
Financial
Healthcare
Industrial
Technology


Weekly total_pnl_usd
1e7
total_pnl_usd
2.4
2.3
2.2
2.1
2.0
1.9
1.8
Week